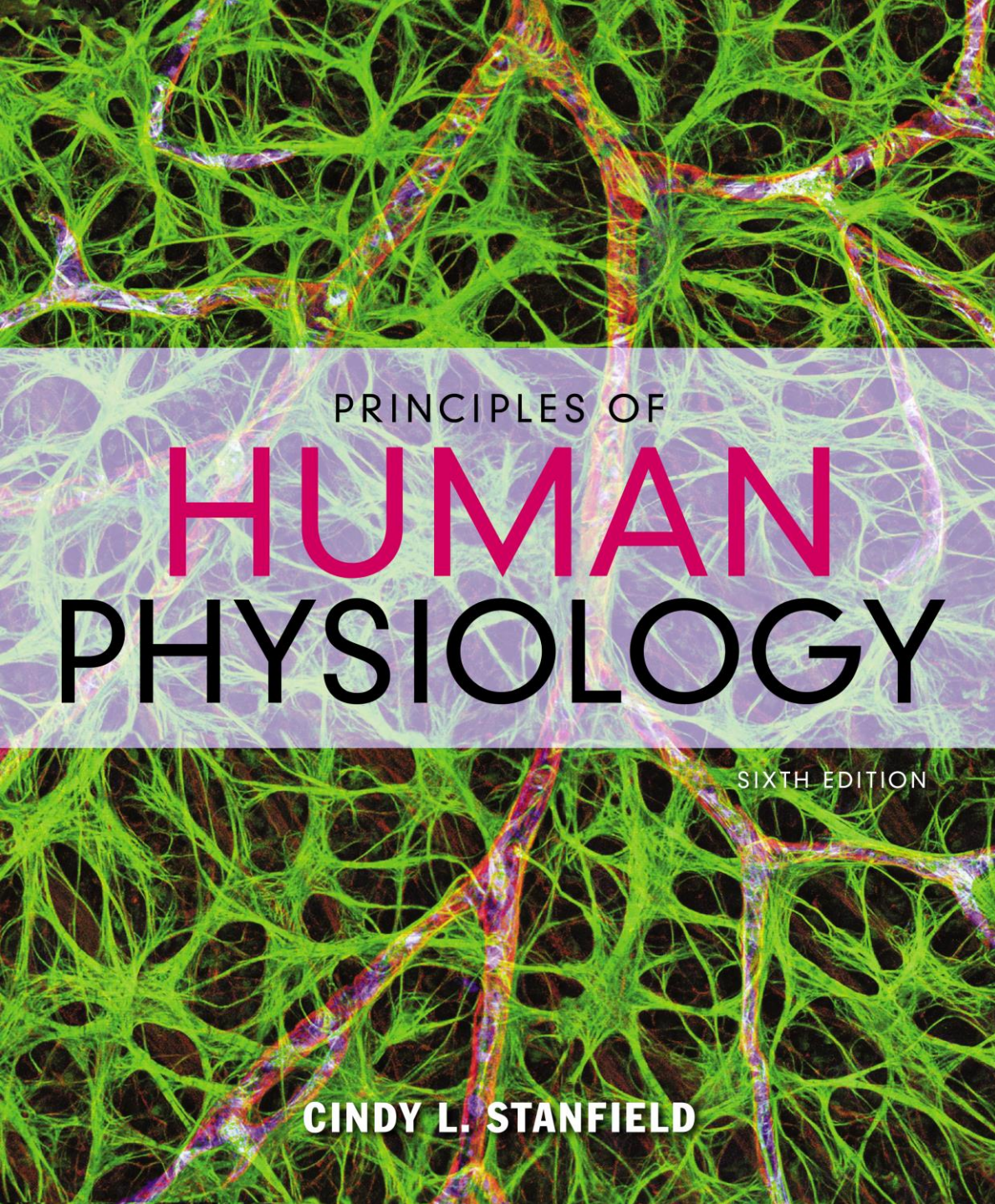




PowerPoint® Lecture
Presentation

PRINCIPLES OF
HUMAN
PHYSIOLOGY

SIXTH EDITION

CINDY L. STANFIELD

CHAPTER **23a**

The Immune
System

Chapter Outline

- 23.1 Anatomy of the Immune System
- 23.2 Pathogens That Activate the Immune Response
- 23.3 Organization of the Body's Defenses
- 23.4 Humoral Immunity
- 23.5 Cell-Mediated Immunity
- 23.6 Immune Responses in Health and Disease

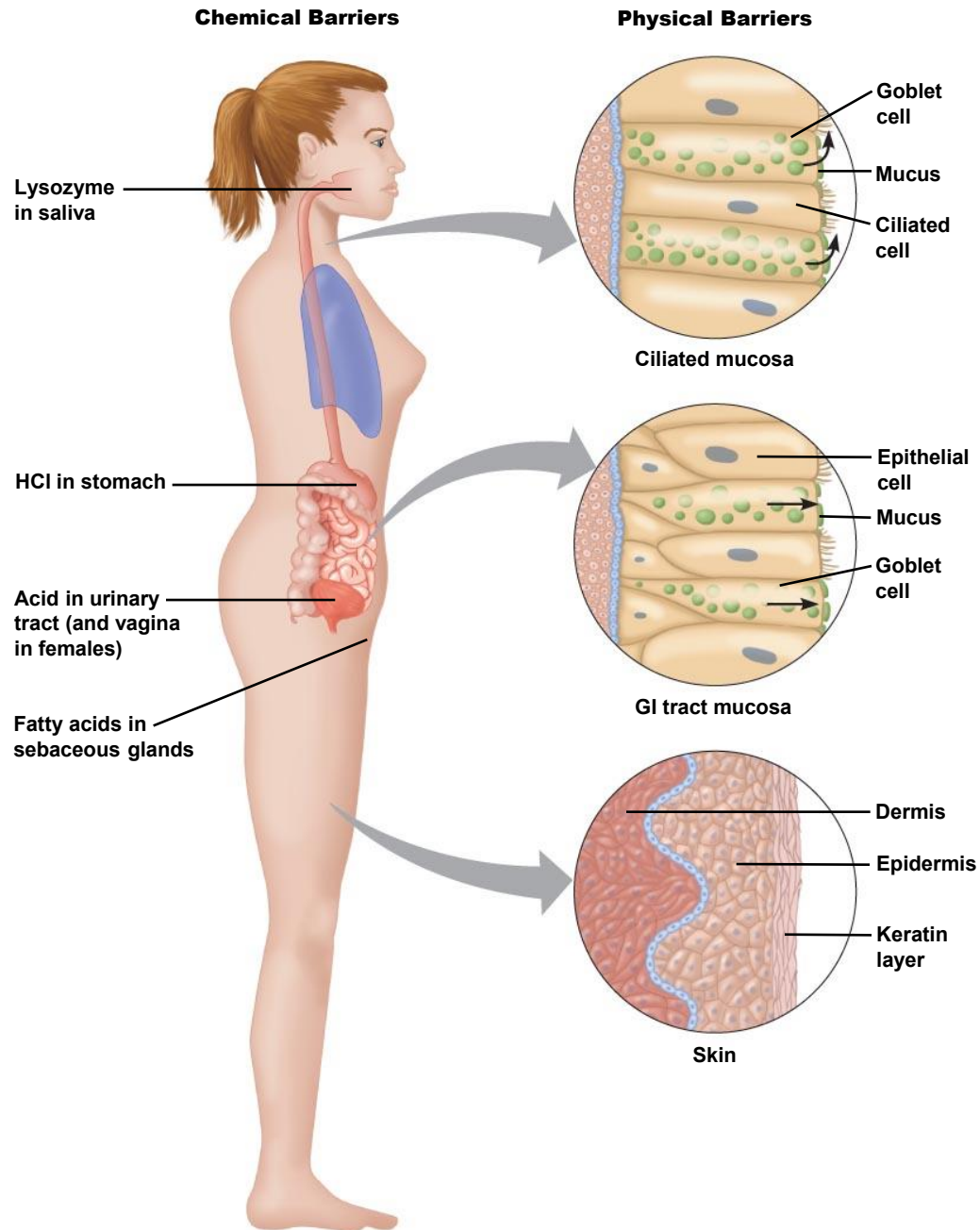
23.1 Anatomy of the Immune System

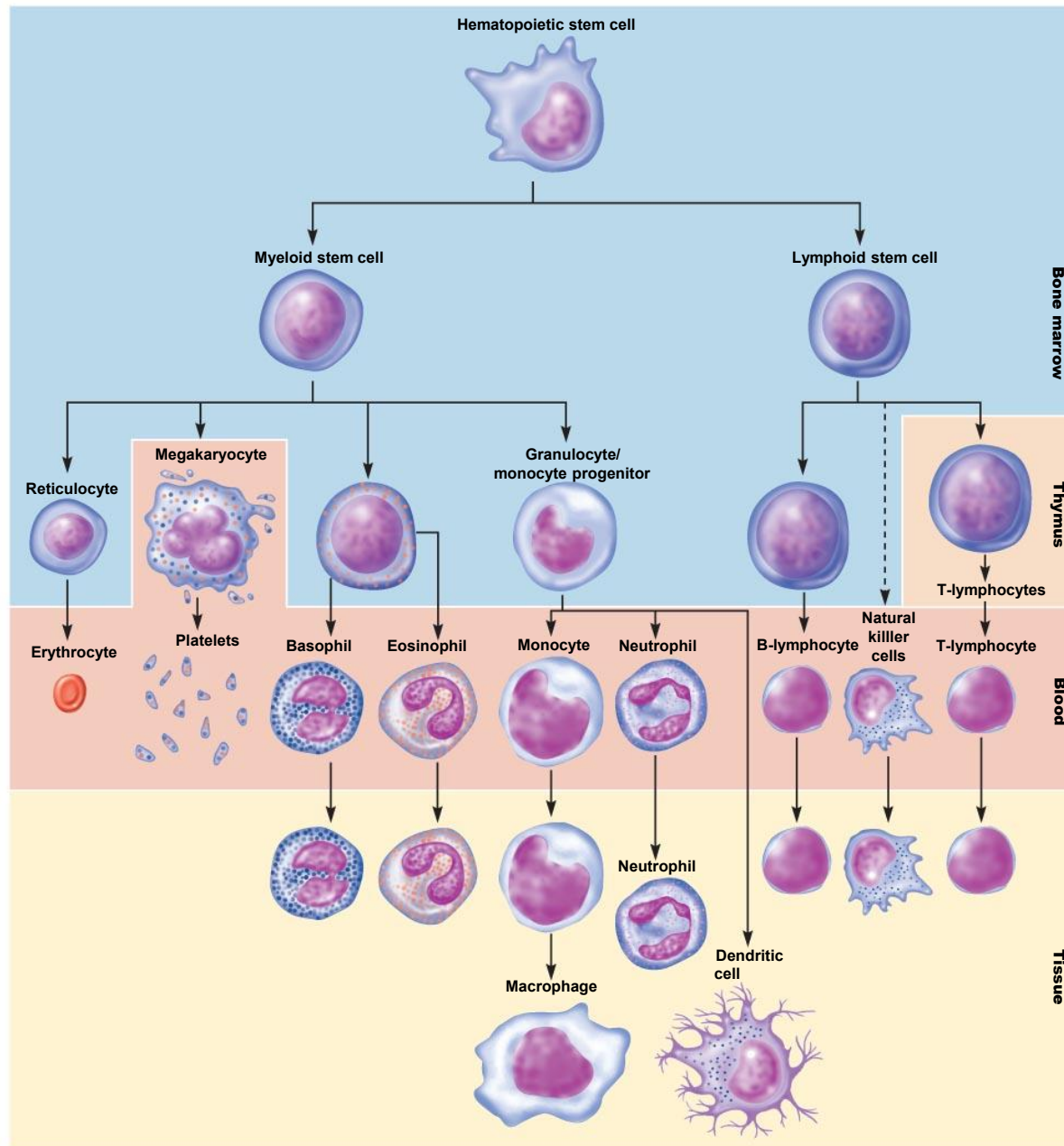
- Physical barriers
 - Skin and mucous membranes
- Leukocytes
 - White blood cells
 - Wide range of immune responses
- Lymphoid tissues
 - Bone marrow, thymus, spleen, lymph nodes, and tonsils
 - Development and function of leukocytes

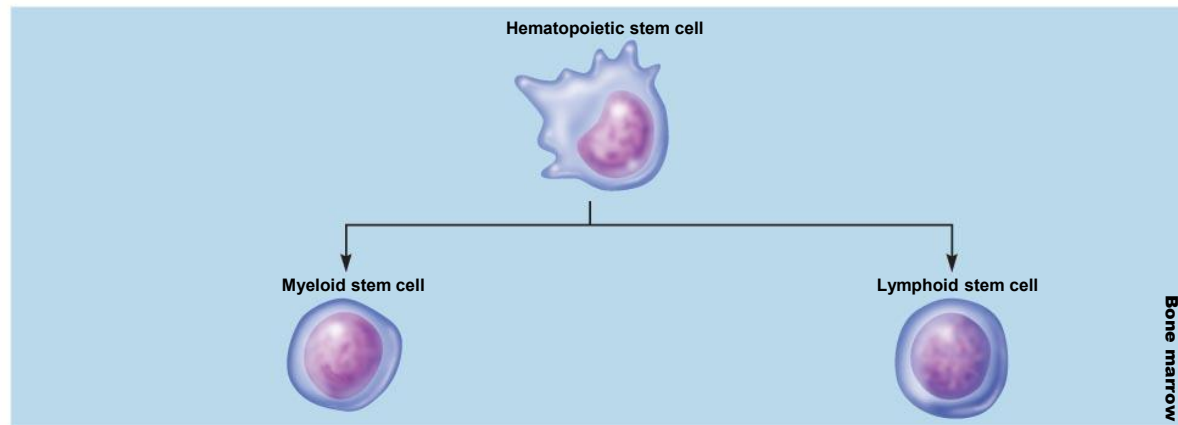
Physical Barriers

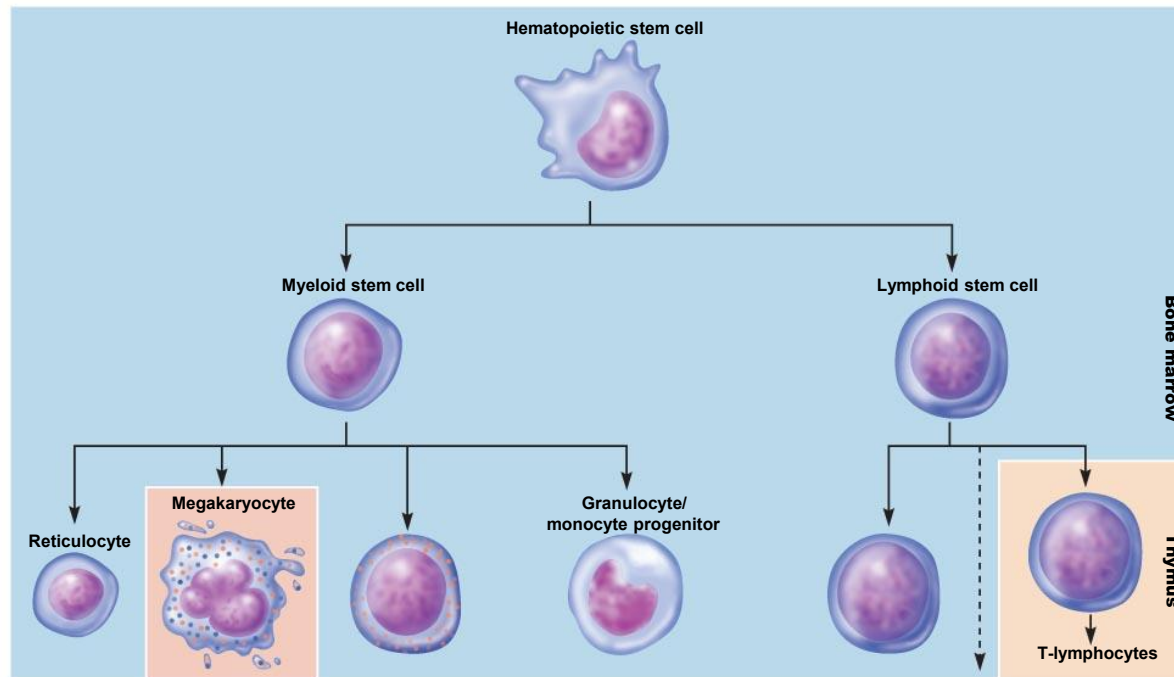
- Skin
 - Epidermis
 - Dermis
 - Sebaceous glands
- Mucous membranes
 - Viscous mucus traps foreign matter

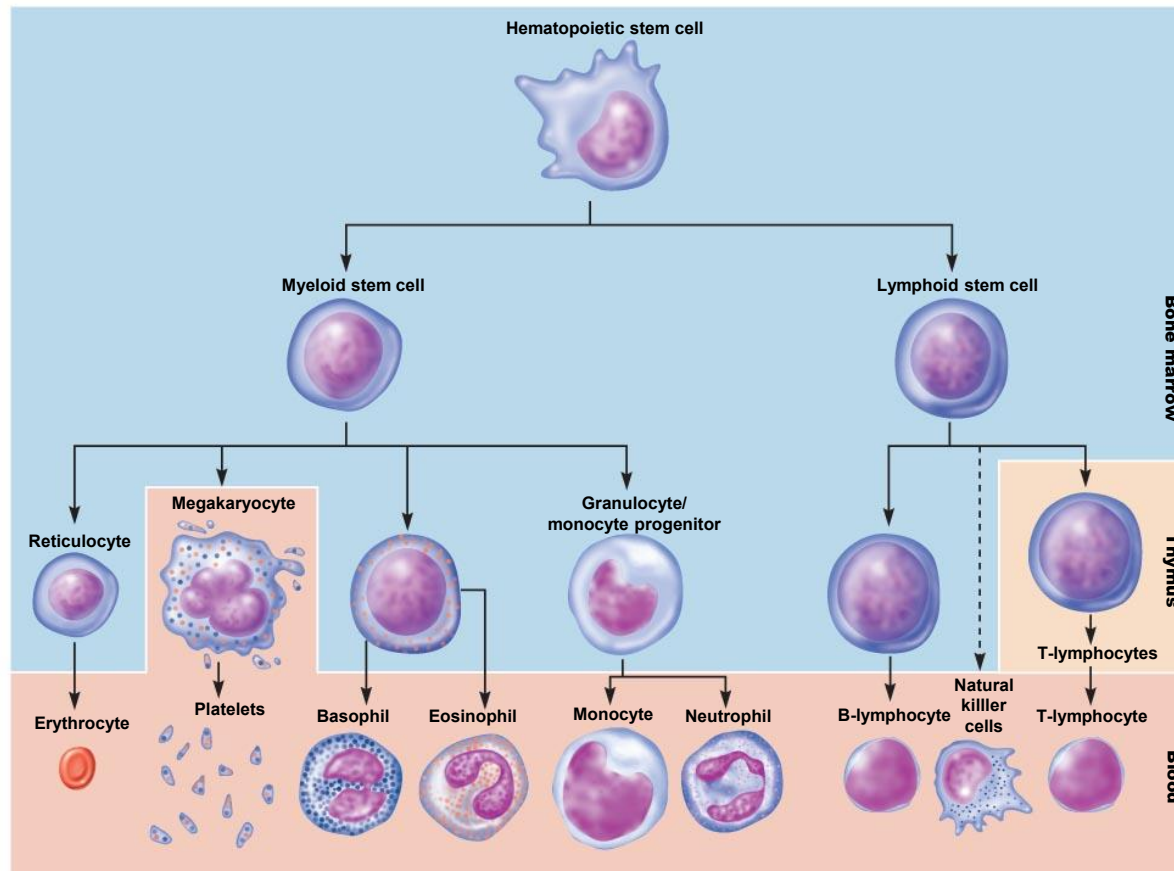
Figure 23.1 Chemical and physical barriers to pathogens.

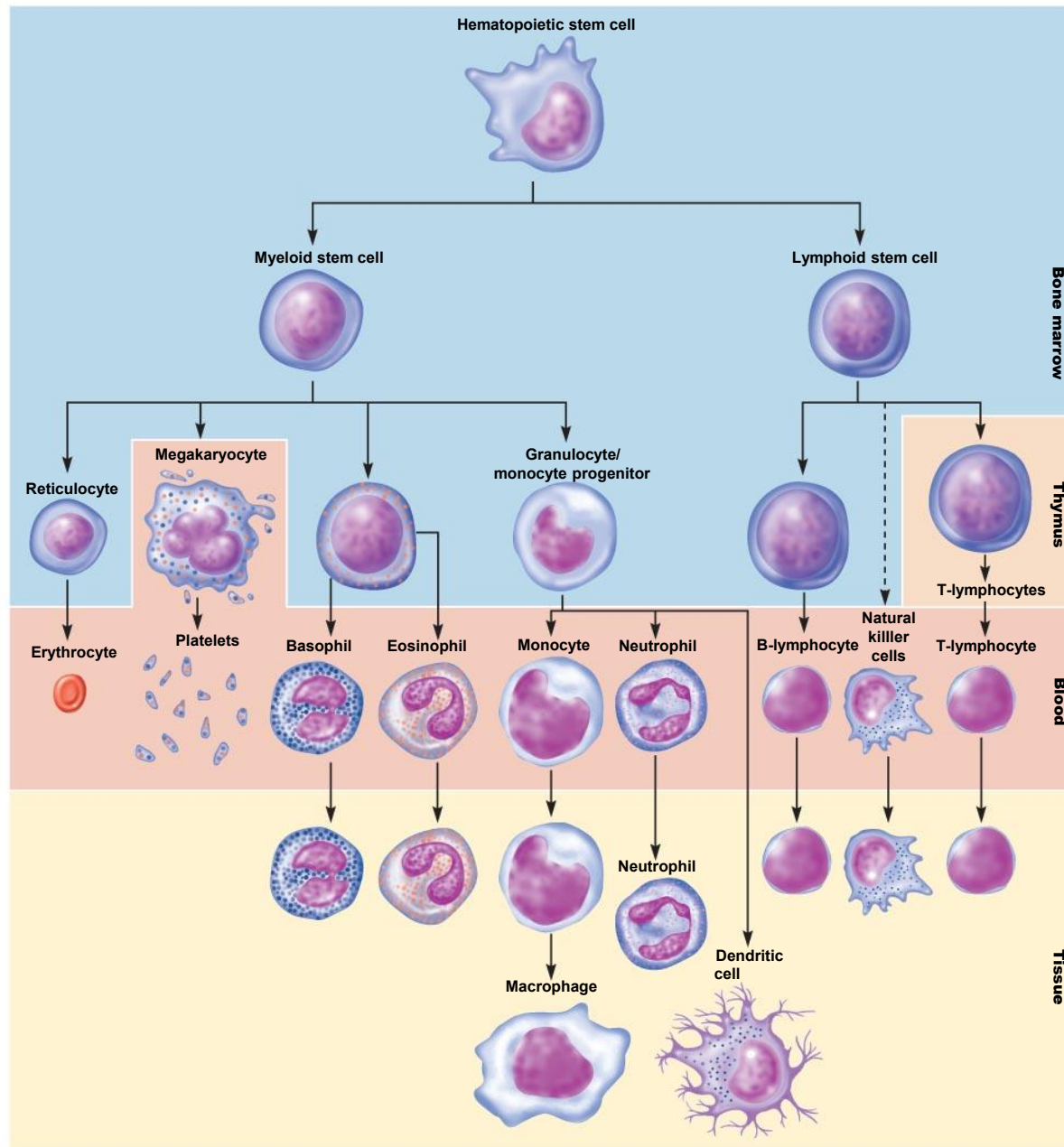












Leukocytes

- Derived from same hematopoietic stem cells in bone marrow
- Granulocytes
 - Neutrophils
 - Eosinophils
 - Basophils
- Agranulocytes
 - Monocytes
 - Lymphocytes

Leukocytes

- Phagocytes
 - Neutrophils (most abundant)
 - Eosinophils
 - Monocytes
 - Macrophages
 - Dendritic cells
- Secrete cytokines
- Circulate in blood 7–10 hours
- Migrate to tissues for a few days
- Increase in number during infections

Leukocytes

- Lymphocytes
 - Account for 20–40% of leukocytes
 - Account for 99% of interstitial fluid cells
 - Three types
 - B lymphocytes (B cells)
 - T lymphocytes (T cells)
 - Null cells

Leukocytes

- B cells
 - Associated with antibodies
 - Contact antigen → plasma cell
 - Plasma cell secretes antibodies (immunoglobulins)
 - Antibodies mark invaders for destruction

Leukocytes

- T cells
 - Directly damage foreign cells
 - Contact infected, mutant, or transplanted cells
 - Develop into cytotoxic T cells that destroy target cells
 - Takes several days
 - Secretory products form pores in the target cell's membrane
 - Lysis

Leukocytes

- Null cells
 - Most are natural killer cells
 - Important defense against viral infections
 - Attack virus-infected cells
 - Cause lysis
 - Fast acting
 - Early immune response

Leukocytes

- Mast cells
 - Immune cells of skin and mucosa
 - Secrete histamine and other substances
- Dendritic cells
 - Similar to macrophages
 - Phagocytes
 - Activate certain types of T cells

Lymphoid Tissues

- Bone marrow
- Thymus
- Spleen
- Lymph nodes
- Tonsils
- Adenoids
- Appendix
- Peyer's patches

Lymphoid Tissues

- Central lymphoid tissue = bone marrow + thymus
- Bone marrow
 - Hematopoietic stem cells: precursors for all blood cells
 - Leukocytes (except T lymphocytes) fully develop in bone marrow
- Thymus
 - T lymphocytes migrate from bone marrow to thymus
 - Reach maturity in thymus

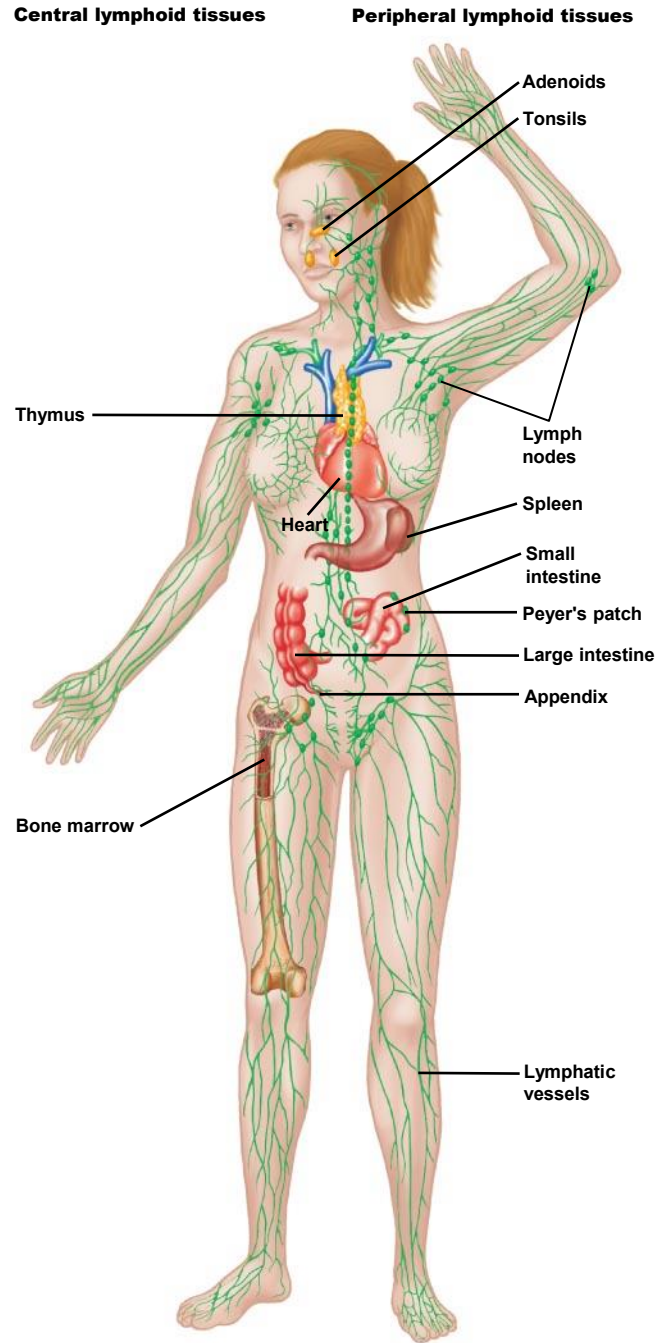
Lymphoid Tissues

- Peripheral lymphoid tissue
 - Spleen, lymph nodes, tonsils, adenoids, appendix, and Peyer's patches
 - Collections of B cells, T cells, and macrophages
 - Trap microorganisms and foreign particles
 - Expose them to leukocytes in high concentrations
 - Spleen and lymph nodes filter blood and lymph

Lymphoid Tissue

- Tonsils and adenoids
 - Trap inhaled particles
- Appendix and Peyer's patches
 - Trap ingested particles

Figure 23.3 Lymphoid tissues.



23.2 Pathogens That Activate the Immune Response

- Four different classes of pathogens
 - Viruses
 - Bacteria
 - Fungi
 - Parasites
- Ability to cause a disease = virulence

Viruses

- Smallest (20–400 nm)
- Single- or double-stranded nucleic acid surrounded by protein coat
- Requires a host
- Uses host's metabolic machinery to reproduce
 - Can double the number of viral organisms hourly
- Target specific cells

Bacteria

- Smallest (1–5 μm)
- Prokaryotic
- Most have single circular chromosome
- Gram-positive or -negative
 - Ability to break up cresyl violet
- Invade tissue
 - Colonization: adhesin
- Release endotoxins or exotoxins

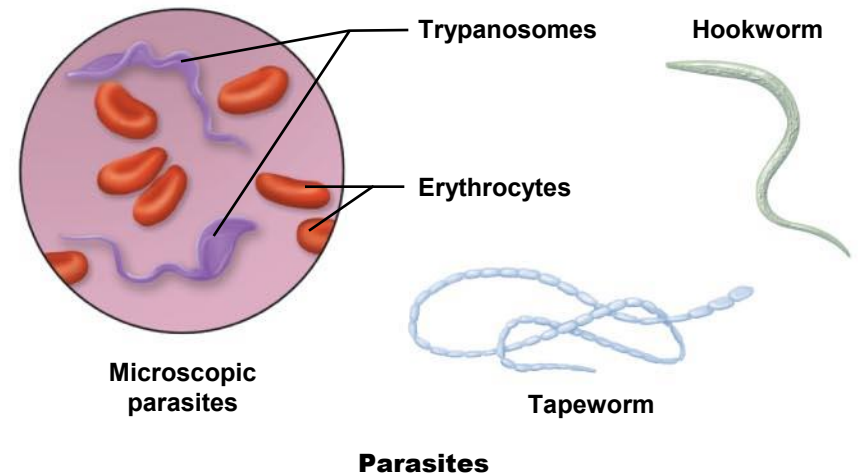
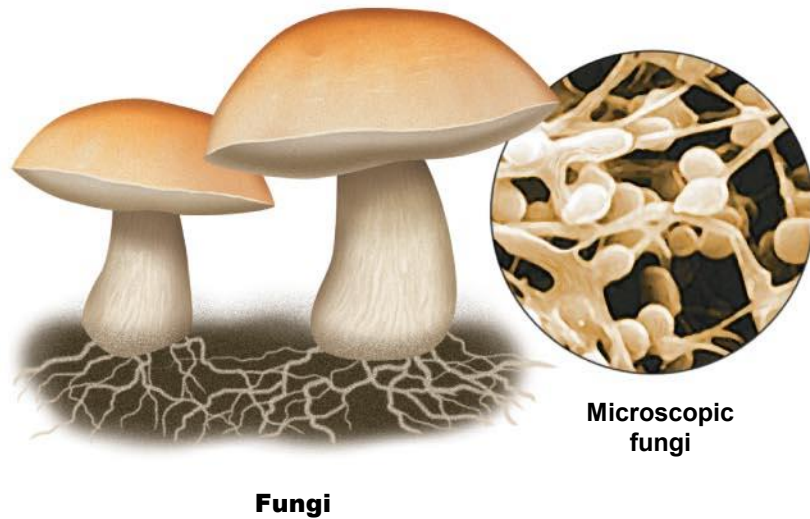
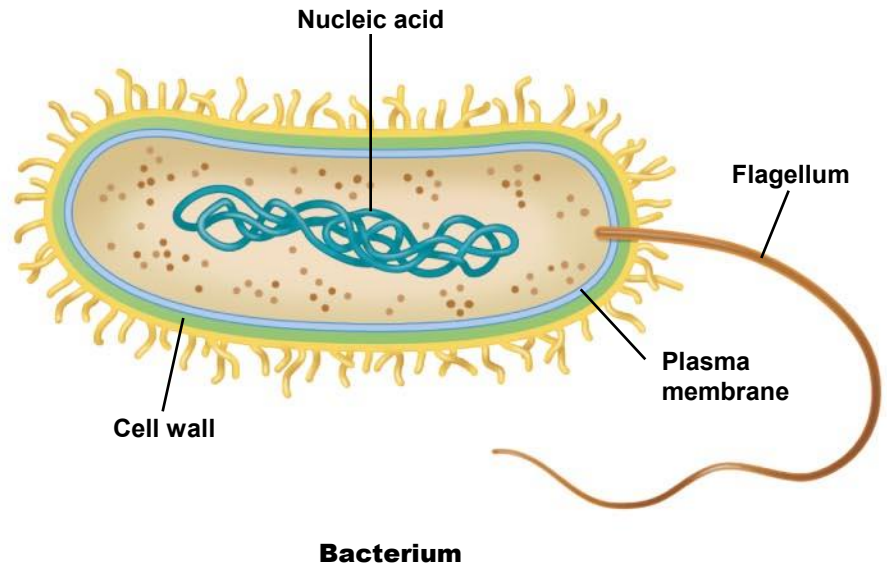
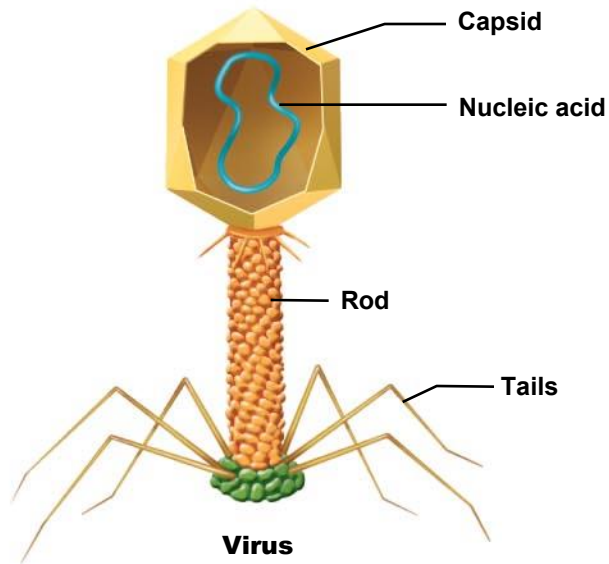
Fungi

- Constitute their own kingdom; neither plants nor animals
- Eukaryotic
- Yeast, molds, and mushrooms
- Absorb nutrients from environment
- Grow slowly

Parasites

- Invading organisms
- Protozoans
 - Example: *Trypanosoma brucei* (sleeping sickness)
- Metazoans
 - Examples: worms, arthropods
- 10 μm to 20 cm in size; most less than 50 μm
- Vector

Figure 23.4 Classes of pathogens: viruses, bacteria, fungi, and parasites.



23.3 Organization of the Body's Defenses

- Nonspecific defenses (innate defenses)
 - Pathogen-associated molecular patterns (PAMPs)
 - Pattern recognition receptors: bind PAMPs
 - Physical barriers
 - Inflammation
 - Interferons
 - Natural killer cells
 - Complement system

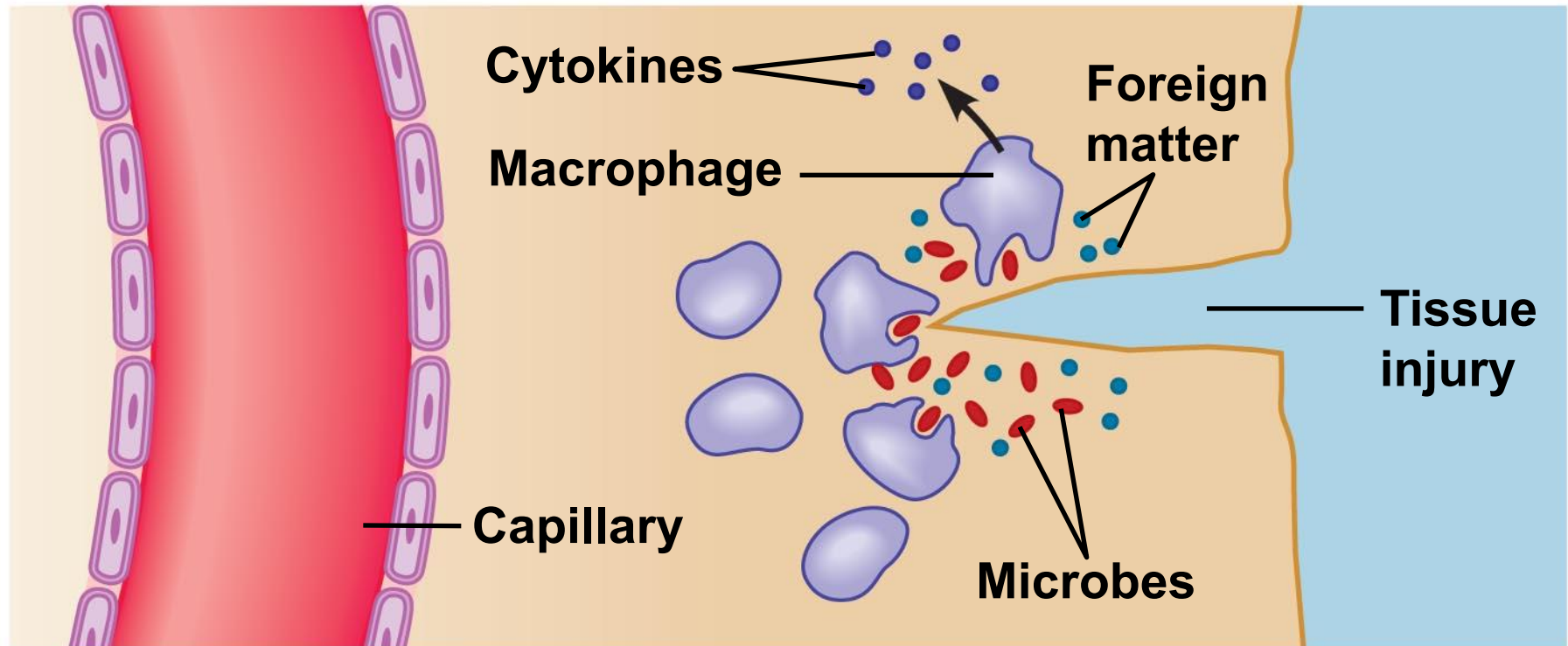
Nonspecific defenses

- Inflammation: series of events causing accumulation of proteins, fluid, and phagocytes in an injured or invaded area
 1. Macrophages engulf debris and foreign matter
 2. Capillaries dilate and become more permeable
 3. Foreign matter is contained
 4. More leukocytes migrate to area
 5. Leukocytes clear infection

Nonspecific Defenses

- Phagocytosis of pathogens
 - Proteins on microbes bound by macrophages
 - Triggers phagocytosis and secretions
 - Secretory products trigger subsequent steps

Figure 23.5a Major events in local inflammation.

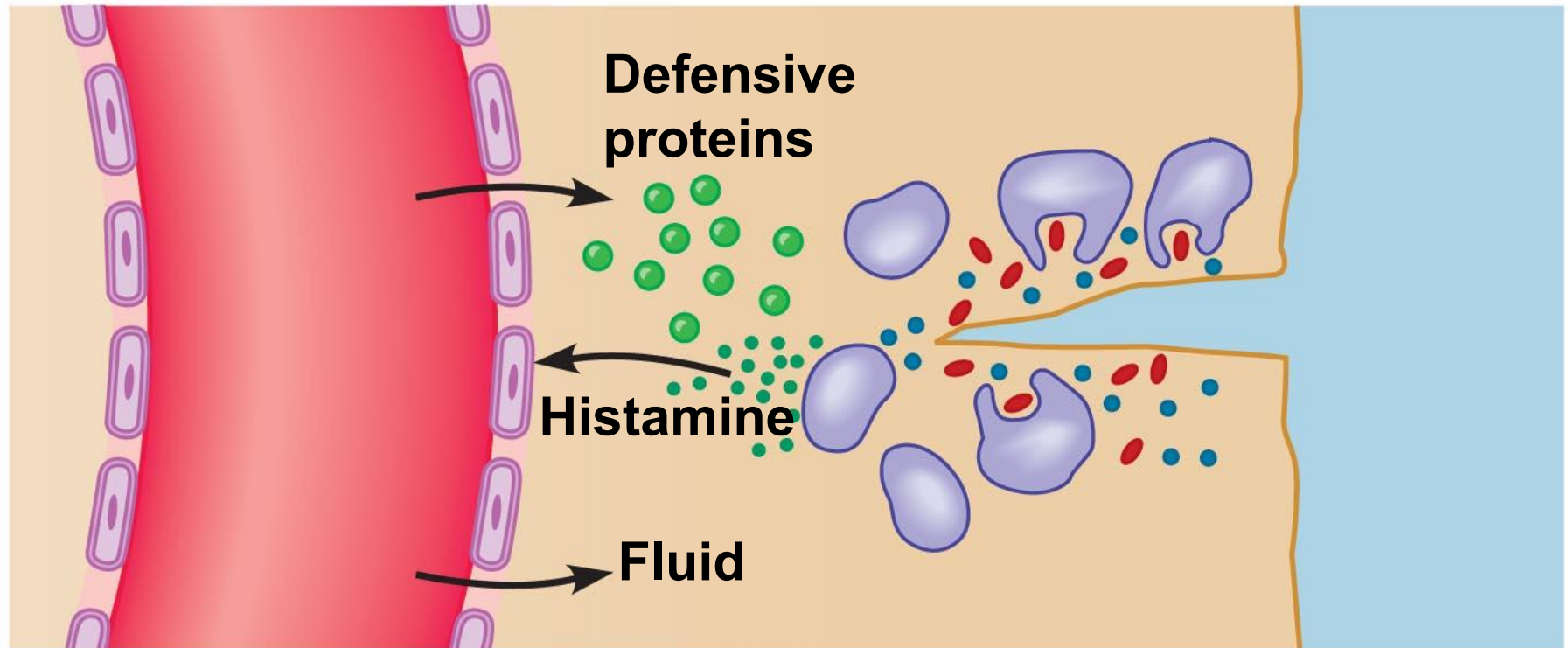


(a) Phagocytosis by nearby macrophages

Nonspecific Defenses

- Dilation and increased permeability in capillaries
 - Damaged mast cells secrete histamine
 - Histamine triggers dilation and increases permeability
 - Result: increased blood flow and increased movement of proteins and cells to injured tissue
- Symptoms of inflammation induced by effects of histamine
 - Redness
 - Heat
 - Edema
 - Pain

Figure 23.5b Major events in local inflammation.

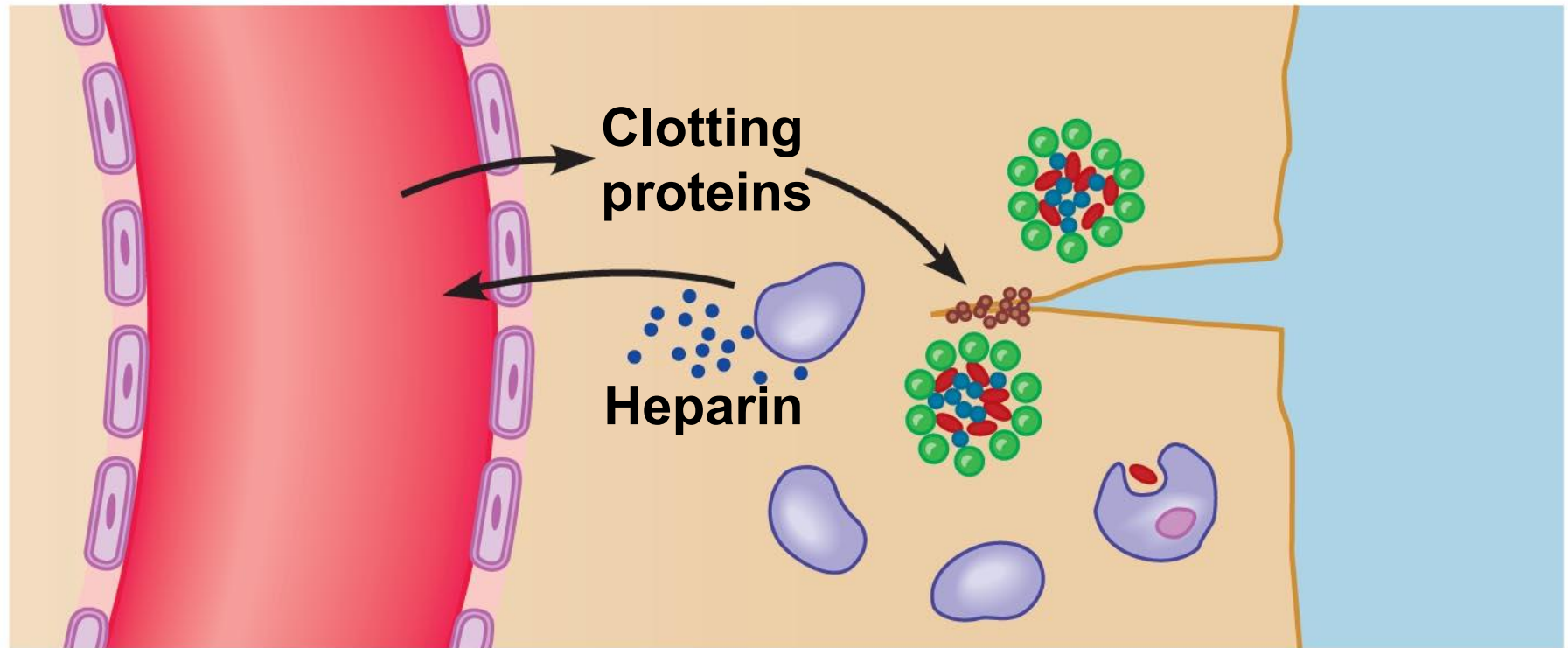


(b) Dilation and increased permeability of capillary

Nonspecific Defenses

- Containment of foreign matter
 - Mast cells and basophils → heparin
 - Heparin prevents clot formation initially so blood cells can access area
- Clotting factors eventually form a clot in tissue, which prevents spread of foreign matter
- Eventually forms a scab

Figure 23.5c Major events in local inflammation.



(c) Containment of bacteria and foreign matter

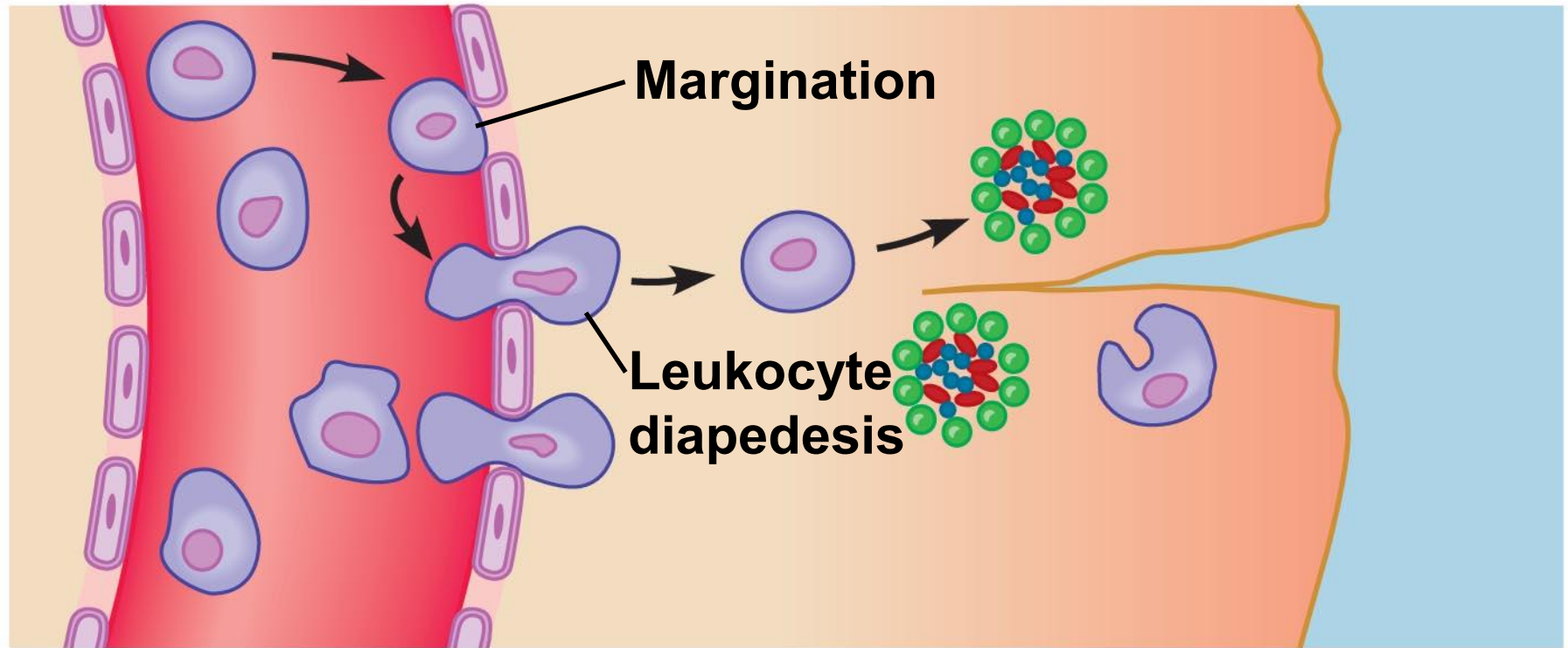
Nonspecific Defenses

- Leukocyte migration and proliferation
 - 1 hour: neutrophil migration
 - 10 hours: monocyte migration → macrophages
- Cells move from blood to tissue triggered by cytokines
 - Margination: move to vasculature wall
 - Attachment: bind to wall
 - Diapedesis: move between endothelial cells
- Movement in tissue to injury: chemotaxis

Nonspecific Defenses

- Leukocyte proliferation
 - Cytokines travel to bone marrow
 - Stimulate production of leukocytes
 - Increase four to five times in circulating numbers

Figure 23.5d Major events in local inflammation.

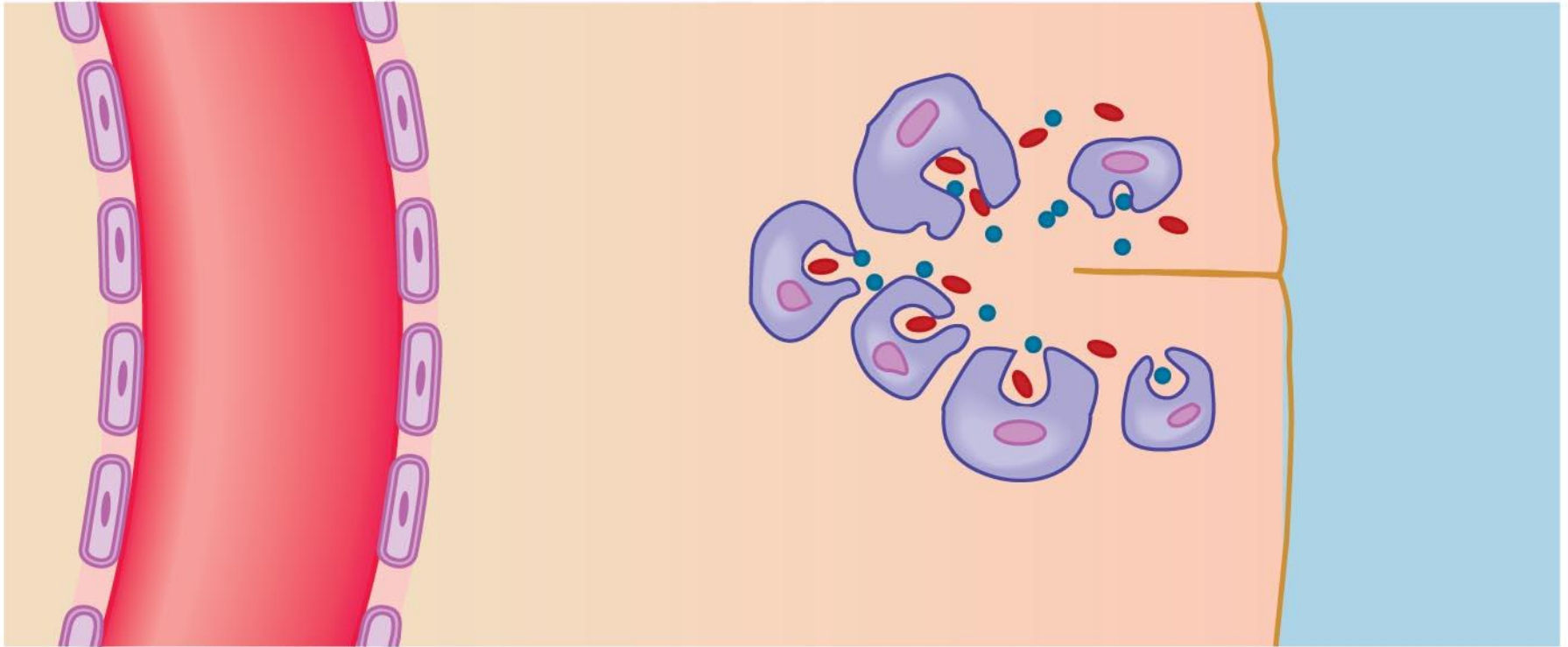


(d) Leukocyte proliferation and migration

Nonspecific Defenses

- Clearing of foreign particles
 - Once foreign particles are present in damaged or invaded tissue, leukocytes must clear the tissue
 - Phagocytosis
 - Secretion of cytokines

Figure 23.5e Major events in local inflammation.



(e) Continued activity of recruited leukocytes

Nonspecific Defenses

- Steps of phagocytosis

1. Attachment

- Specific: damaged cells or protein-targeted cells
- Proteins = opsonins

2. Internalization

- Takes approximately 0.01 sec
- Phagosome + lysosome → secondary lysosome

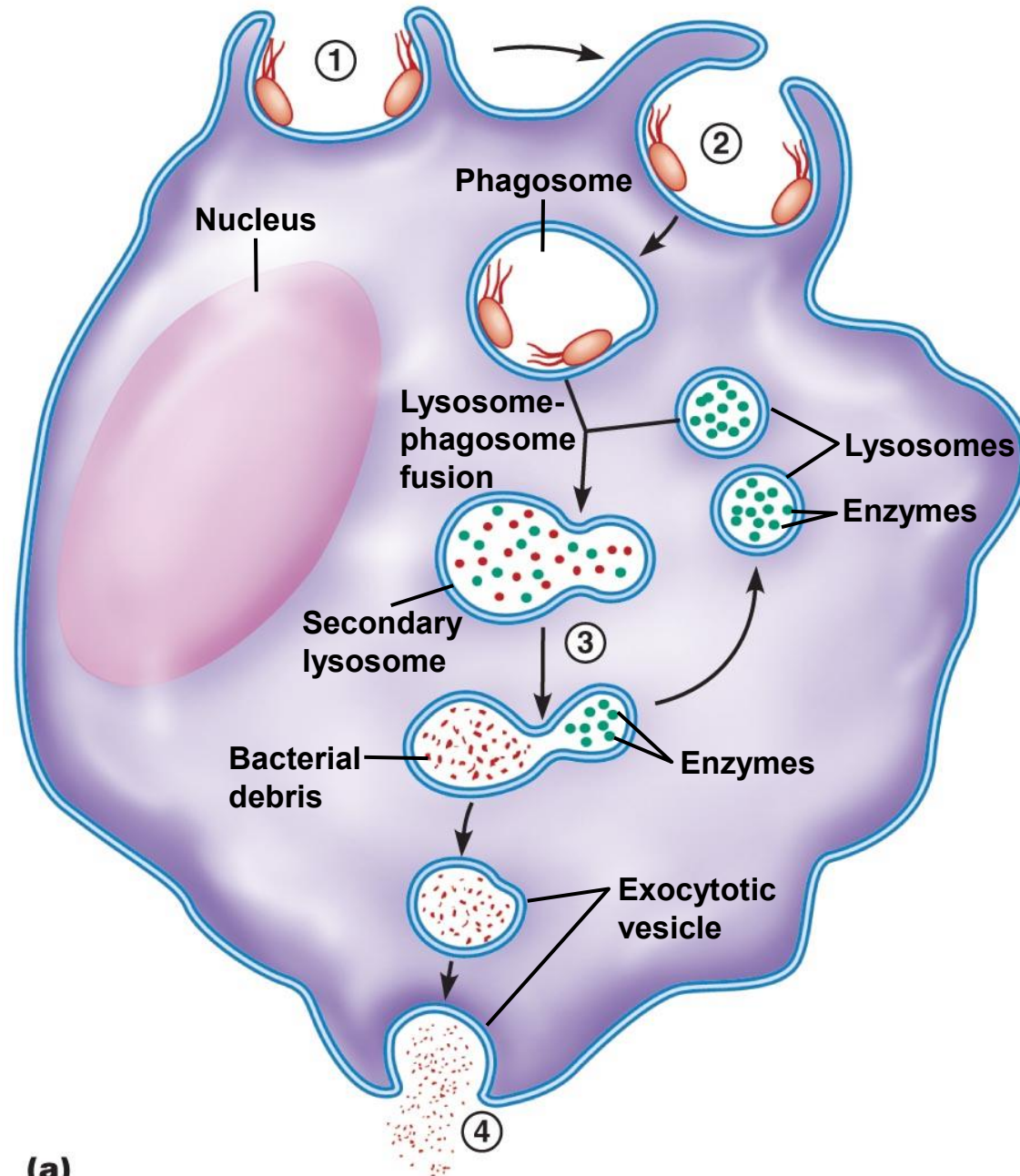
3. Degradation

- Lysosome enzymes degrade phagocytosed product

4. Exocytosis

- Elimination of some degradation products

Figure 23.6a Phagocytosis.



(a)

Figure 23.6b Phagocytosis.



(b)

Nonspecific Defenses

- Secretion of cytokines by phagocytes
 - Trigger: attachment of foreign particle to phagocyte
 - Secretory products of macrophages
 - Interleukins: communicate between leukocytes
 - IL-1
 - IL-6
 - TNF- α

Nonspecific Defenses

- Actions of interleukins and $\text{TNF-}\alpha$
 - Bone marrow: produces and secretes more leukocytes
 - Act as endogenous pyrogens
 - Stimulate fever via hypothalamus
 - Hypothalamus increases body temperature via prostaglandins
 - Liver: acute-phase proteins
 - IL-1 stimulates proliferation and differentiation of B and T lymphocytes
 - B cells → antibody production
 - T cells → cell-mediated immunity

Nonspecific Defenses

- Interferons

- Proteins secreted by leukocytes and virus-infected cells
- Induce virus resistance in neighboring cells
- Viral infection
- RNA invades host cell, using that cell's replication machinery to reproduce
- Viral nucleic acid stimulates host cell to secrete interferon- α and interferon- β
 - Kills host cells
 - Induces neighboring cells to resist viral infection

Nonspecific Defenses

- Interferon- γ
 - Secreted from active T cells and NK cells
 - Inhibits viral replication
 - Enhances phagocytosis by macrophages
 - Boosts antibody production in B cells
 - Helps activate NK cells and cytotoxic T cells
 - Inhibits cell division—defense against cancer

Nonspecific Defenses

- Natural killer cells
 - Recognize abnormal or infected cells
 - Cause lysis of pathogen by secreting perforins
 - Can attack virus-infected cells without identifying virus
 - Early defense against viral infections
 - Part of specific immune response
 - Actions are enhanced by interleukin-2 produced by T cells as well as by antibodies produced by plasma cells

Nonspecific Defenses

- Complement system
 - Plasma proteins that lyse foreign cells, especially bacteria
 - Part of response to antibodies (specific immunity) in addition to nonspecific response
 - Approximately 30 proteins participate in the response cascade, resulting in MAC reaching the surface of bacteria
 - MAC pierces the bacterial membrane, causing lysis

Nonspecific Defenses

- Activation of complement system
 - Classical pathway
 - Binding to antibodies attached to bacteria
 - Part of specific defense mechanisms
 - Lectin pathway
 - Binding to mannose-containing carbohydrates on bacteria or viruses
 - Alternative pathway
 - Binding to carbohydrates on bacterial cell walls
 - Part of nonspecific defense mechanisms

Figure 23.7 Complement activation and actions.

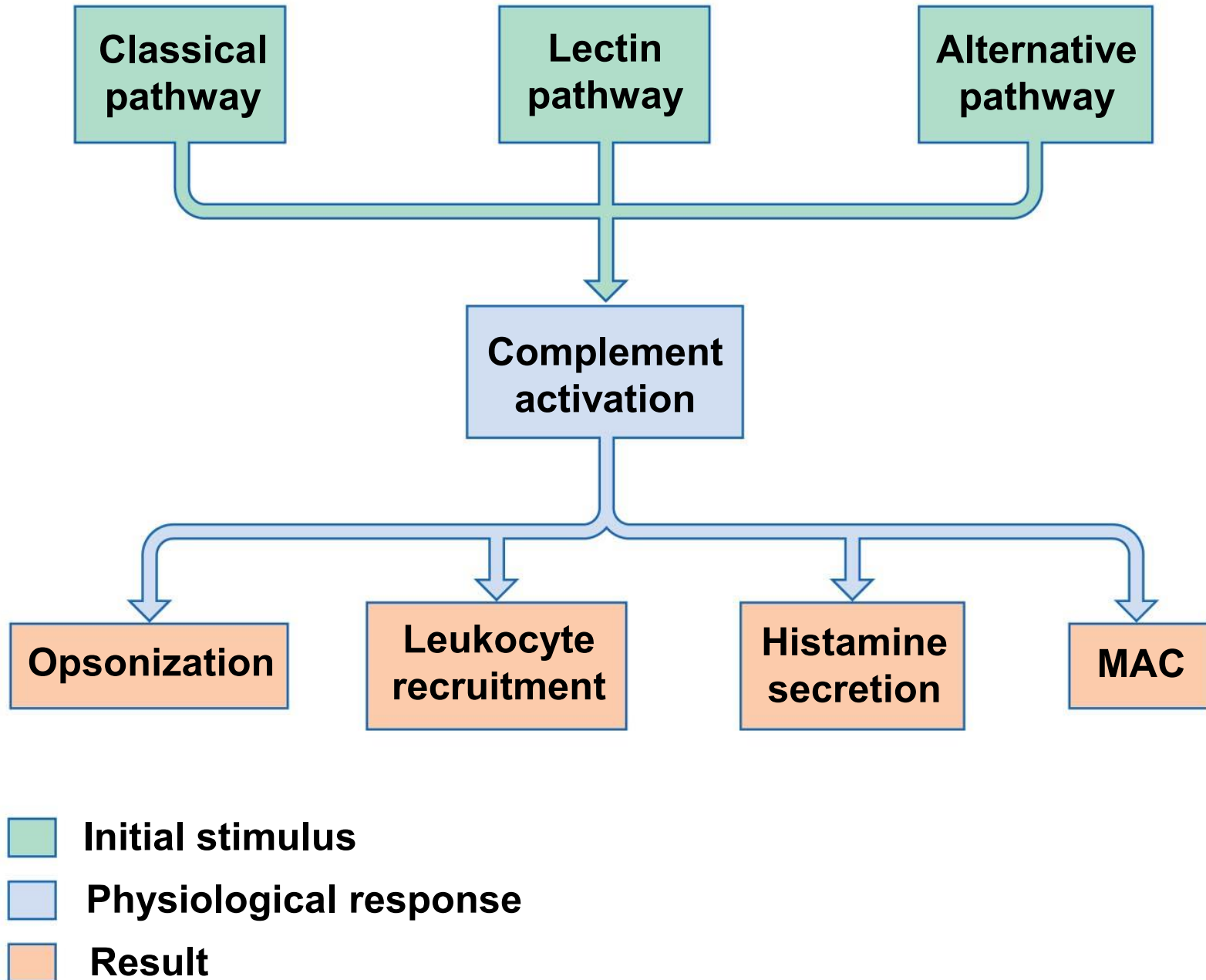
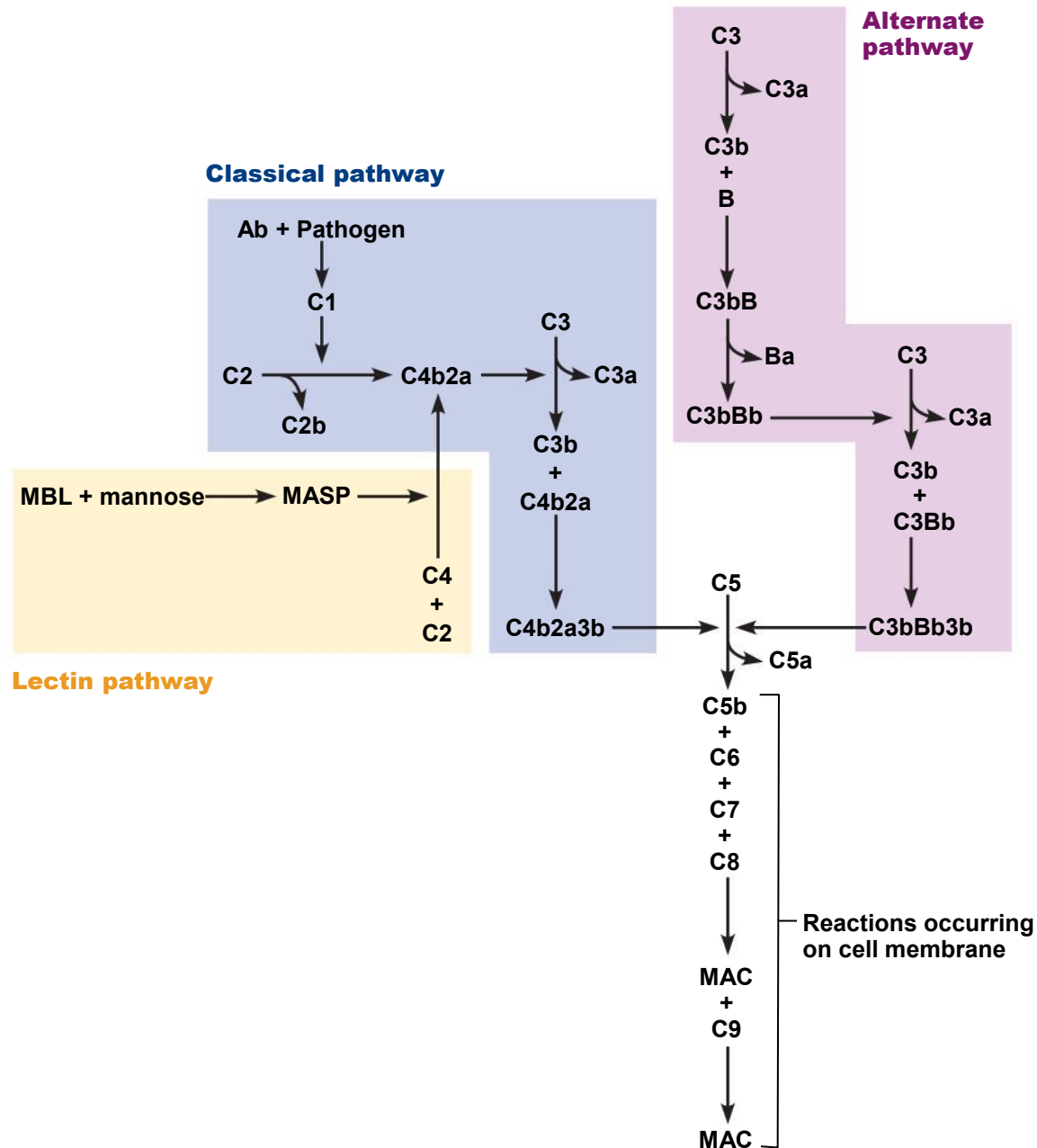


Figure 23.8 Complement pathways.



Nonspecific Defenses

- Complement protein functions
 - Form MAC to lyse cell
 - Chemotaxis
 - Trigger histamine release from mast cells
 - Act as opsonins

Figure 23.9 Formation of the MAC complex and cell lysis.

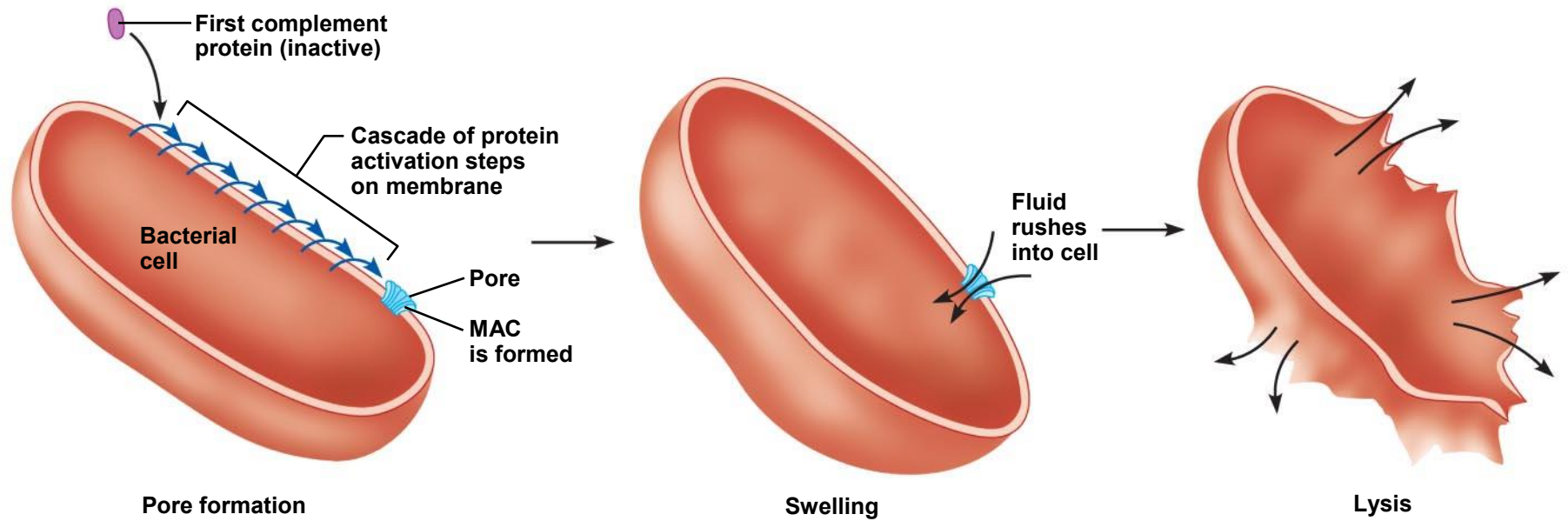
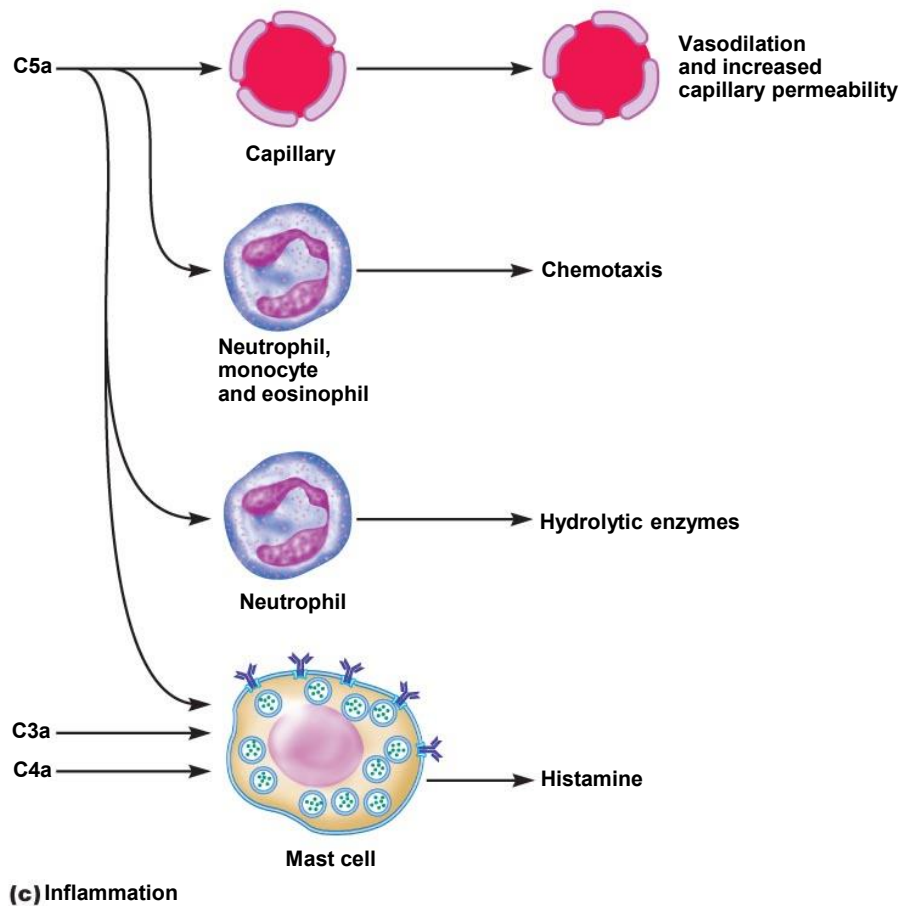
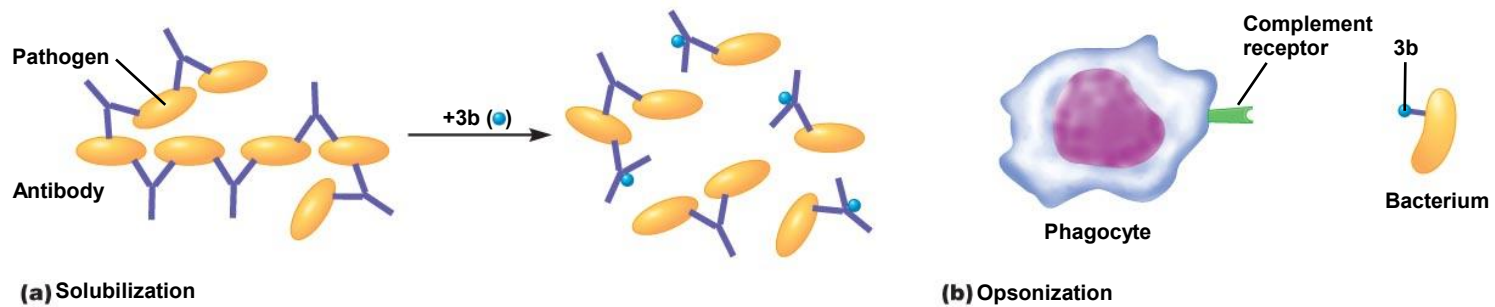


Figure 23.10 Actions of complement proteins independent of MAC.



Specific Defenses: Immune Responses

- Humoral immunity
 - B cell mediated
 - Involves secretion of antibodies by plasma cells
 - Defend against bacteria, toxins, and viruses in body fluids
- Cell-mediated immunity
 - T cell mediated
 - Involves lysis of cells by cytotoxic T cells
 - Defend against bacteria and viruses in body cells
 - Part of reaction to transplants and cancer cells

Specific Defenses: Immune Responses

- Features of specific immune response
 - Specificity
 - Diversity
 - Memory
 - Self-tolerance

Specific Defenses: Immune Responses

- Specificity
 - Antigens (antibody generators) confer specificity
 - Complex proteins and polysaccharides
 - Part of foreign invaders and tumor cells
 - Epitopes: recognition sites for B or T cells
 - B cell and T cell specificity
 - Antigen receptors recognize certain antigens only
 - B cells: membrane antibodies
 - T cells: T cell receptors

Specific Defenses: Immune Responses

- Antibody structure
 - Two heavy chains
 - Two light chains
 - Constant region: same within a class of antibodies
 - Variable region: differs for different antigens, gives specificity to antigen-binding site
 - Two antigen-binding sites

Figure 23.11 The basis of antigen-antibody specificity.

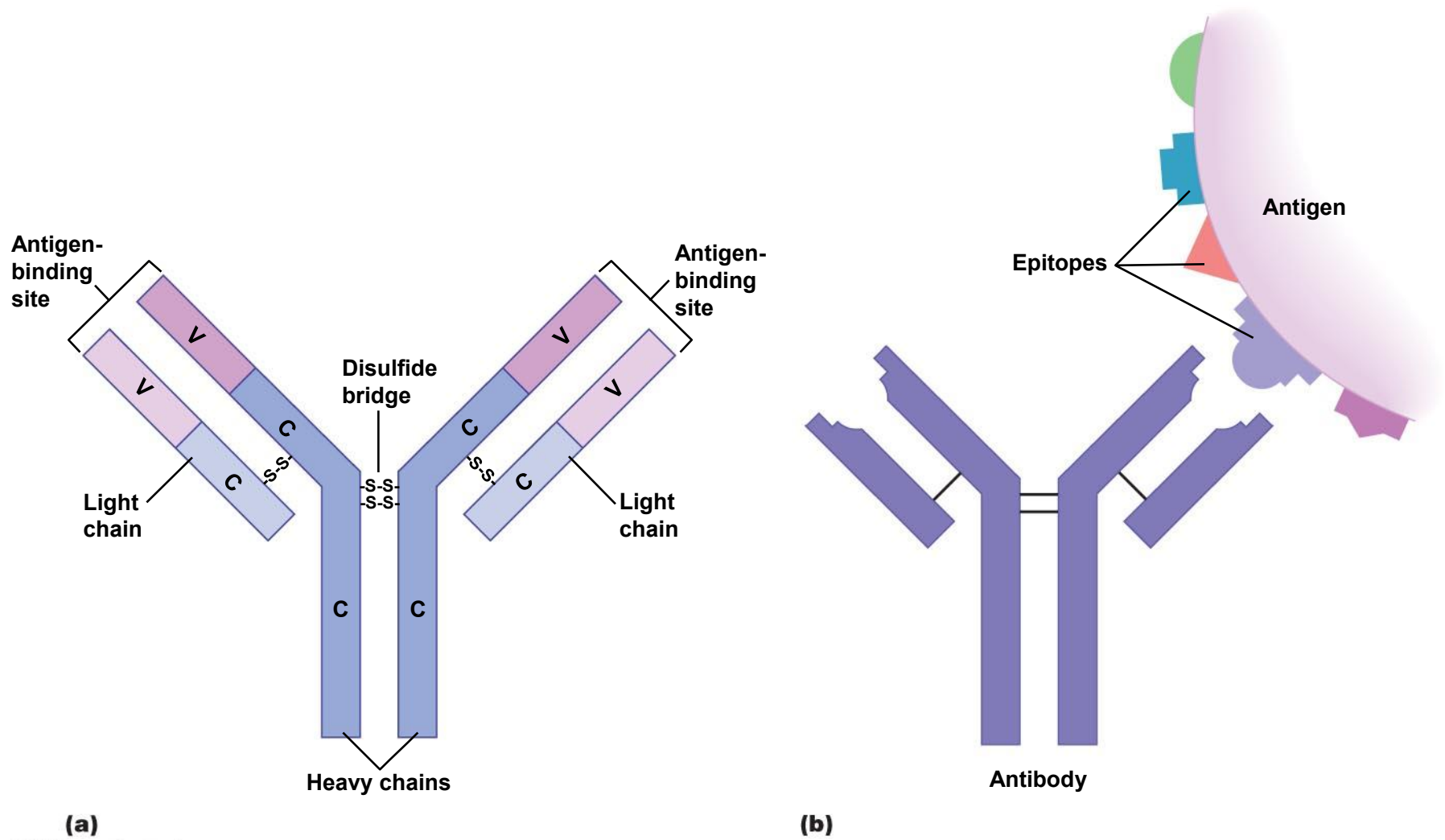
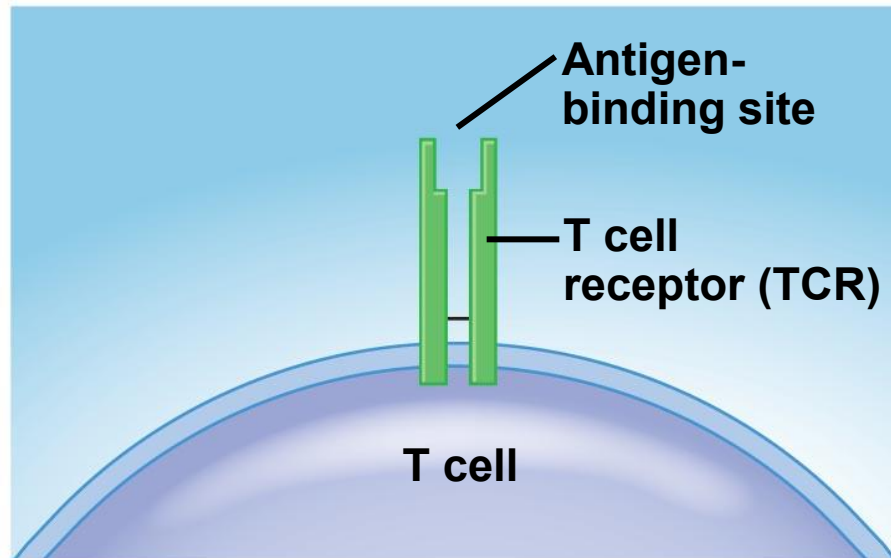
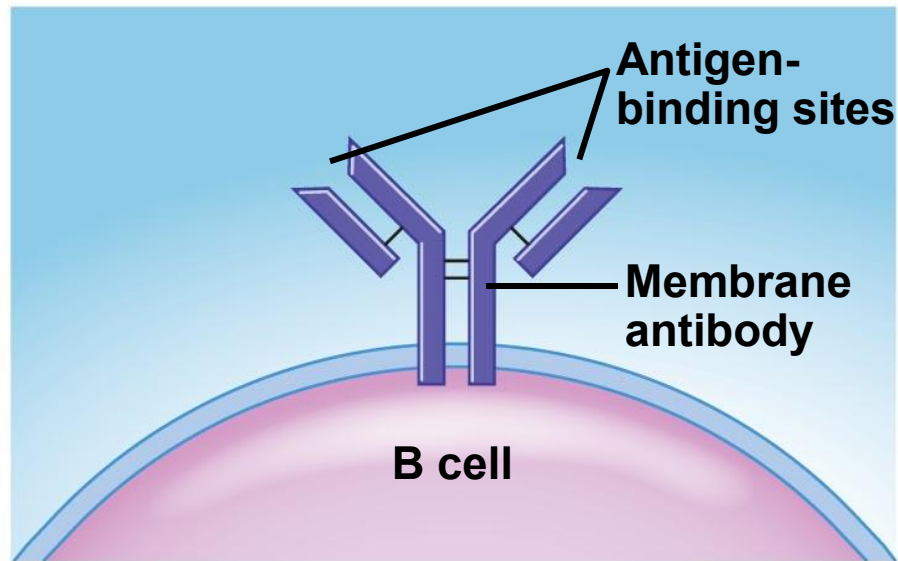


Figure 23.12 B cell and T cell antigen receptors.



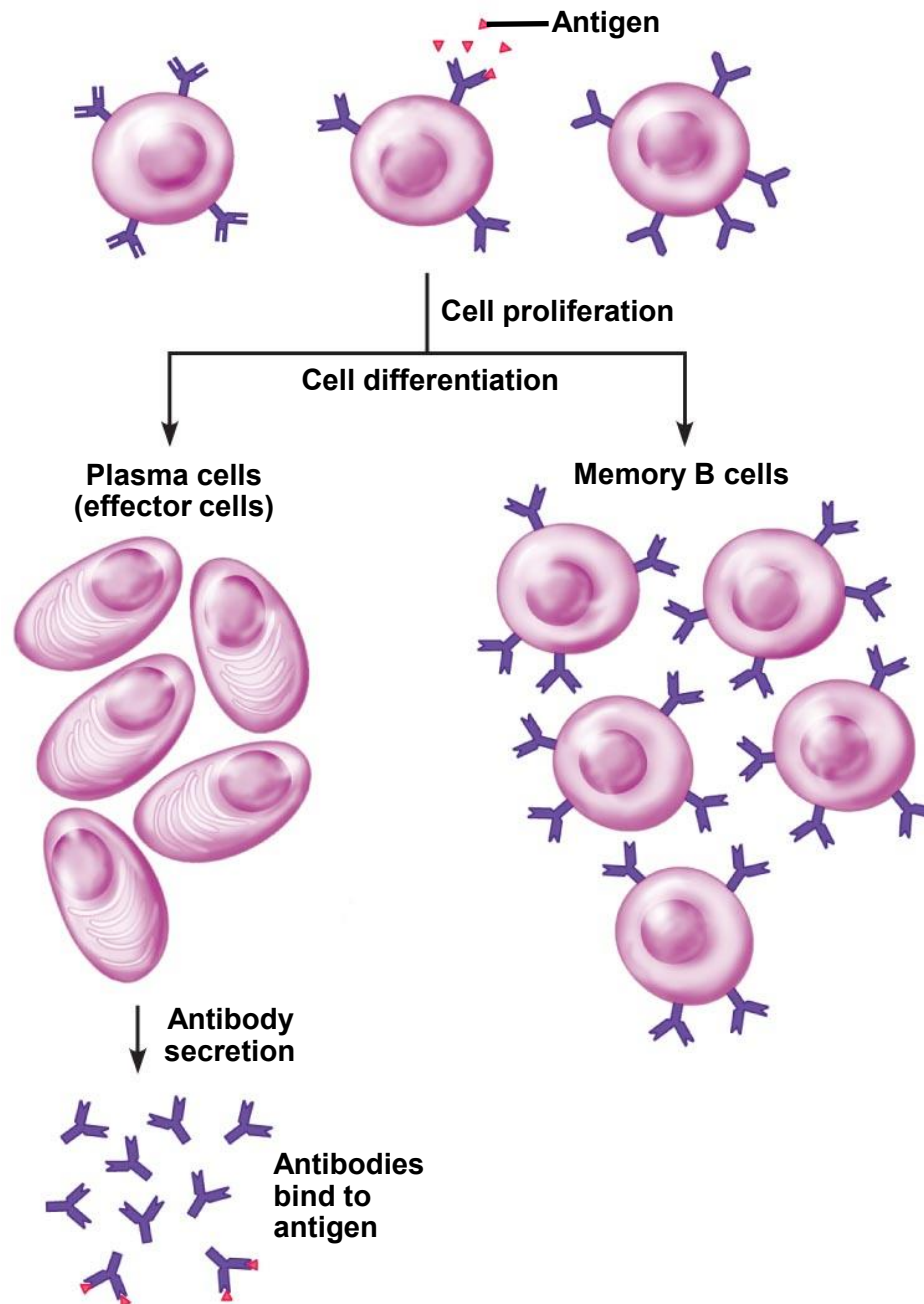
Specific Defenses: Immune Responses

- Diversity
 - Specificity means that a B or T cell can recognize only a few of many antigens; determined genetically
 - Diversity in B- and T-cell antigen receptor types allows the immune system to recognize millions of antigens
 - Antigen interacts with specific lymphocyte
 - Induces those lymphocytes to proliferate and differentiate
 - Activation of lymphocytes = clonal selection

Specific Defenses: Immune Responses

- Two types of clones
 - Effector cells
 - Combat the antigen that stimulated their production
 - Short lived
 - Memory cells
 - Long lived
 - Antigen specific

Figure 23.13 Clonal selection in B lymphocytes.



Specific Defenses: Immune Responses

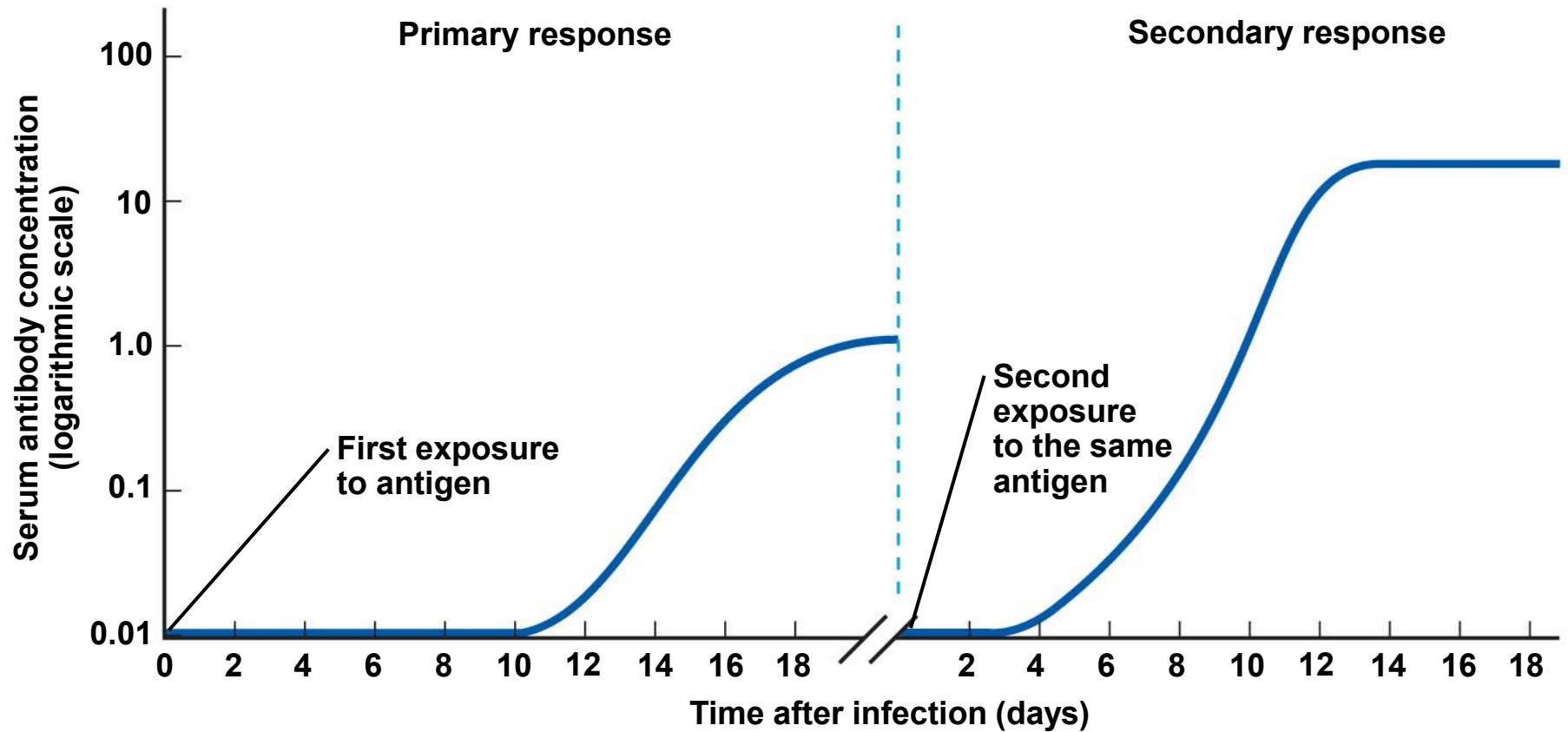
- Memory

- First exposure to antigen
- Takes 10–17 days to occur after exposure
- Symptoms of illness occur during these days
- Antigen-selected B and T cells proliferate and differentiate into effector cells
 - Plasma cells
 - Cytotoxic T cells

Specific Defenses: Immune Responses

- Secondary immune response
 - Subsequent exposures to antigen
 - Takes 2–7 days to occur
 - Response has greater magnitude and is more prolonged
 - Occurs due to presence of memory cells

Figure 23.14 Primary and secondary responses to a given antigen.



Specific Defenses: Immune Responses

- Self-tolerance
 - B and T cells do not attack normal cells of body
 - As cells develop in bone marrow and thymus, any that have antigen receptors against normal body cells are destroyed by apoptosis (programmed cell death)
 - Autoimmune diseases caused by failure of self-tolerance